MTH1W

Slope is a Rate of Change

A **unit** is a quantity used to measure something. (e.g. seconds, years, metres, kilograms, dollars)

A **rate of change** is the change in one thing relative to the change in another.
- km/h, dollars each month, goals per game

Slope is a measure of the rate of change (steepness of a line)
- It is the ratio of the rise (up) to the run (right).

Practice

Identify each as a rate or not.

Measurement	Is it a Rate? Circle one.
You drove 340 km to the cottage	Yes No
You drove to the cottage at 89 km/h	Yes No
The Raptors scored 113 points tonight	Yes No
The Raptors average 110.6 points per game.	Yes No
I earned a $200 bonus at work.	Yes No
I earned a $100 bonus for every extra car I sold.	Yes No
2.1 kg/L	Yes No
0.6 cm/week	Yes No
8 slices of pizza	Yes No
$12 monthly fee	Yes No
7.6 L/100 km	Yes No
$12 donation	Yes No

$Slope = \frac{rise}{run}$ is often how we show the rate of change in math.
We often use *m* for slope (no one knows why), but it is interesting that the French word *monter* = to climb

$m = \frac{rise}{run} = \frac{change\ in\ y}{change\ in\ x}$

Practice

1. You drove a total of 175 km in 2.5 hours. What was your rate?

2. My bank account earned $900 in interest over the last year. How much did I earn each month?

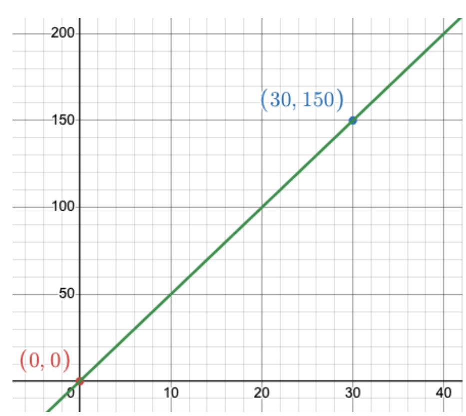
3. A car traveled 265 km on a 30 L fill-up of gasoline.

a) How many kilometers does the car travel on a single Litre?

b) How many liters are burned per km traveled?

4. Toronto Hydro charges you $26.21 in exchange for 126 kWh of electricity.
How much are you paying for every kWh?

5. What is the slope of the line shown on the right?



Homework

1. An empty swimming pool is being filled.
The volume of water *y* in the pool after *x* minutes increases **linearly**

The swimming pool contains 500 L of water after 5 minutes.

a) What is the **initial value** (with units)?

b) What is the **slope** (with units)?

c) Represent this situation in the following THREE ways.
(No title or axis labels needed on the graph here)

<table>
<tr><th>Table of Values</th><th>Graph</th><th>Equation</th></tr>
<tr><td>
<table>
<tr><th>x (minutes)</th><th>y (liters)</th></tr>
<tr><td>0</td><td></td></tr>
<tr><td>5</td><td></td></tr>
<tr><td>10</td><td></td></tr>
<tr><td>15</td><td></td></tr>
<tr><td>20</td><td></td></tr>
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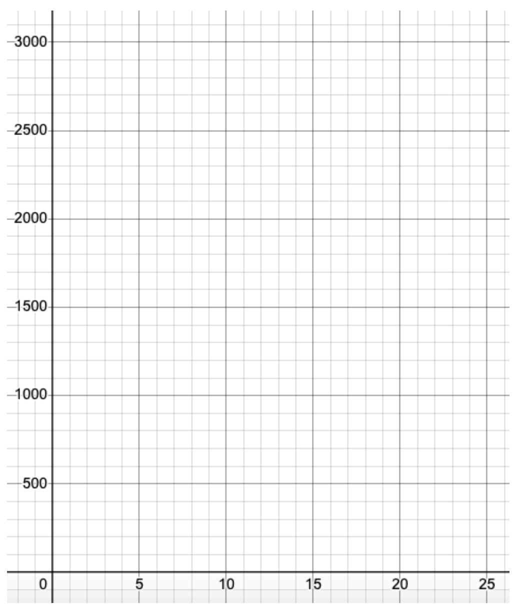

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d) Use the **graph** to estimate the volume of water in the pool after 25 minutes.

e) Use the **equation** to **calculate** the volume of water in the pool after 25 minutes.

2. The cost, y, in dollars, of building a concrete sidewalk varies linearly with its length, x, in meters. There is no "start-up" fee.

A 200-m sidewalk costs $4500.

a) What is the **initial value** (with units)?

b) What is the **slope** (with units)?

c) Represent this situation in the following THREE ways.
(No title or axis labels needed on the graph here)

<table>
<tr><th>Table of Values</th><th>Graph</th><th>Equation</th></tr>
<tr>
<td>
<table>
<tr><th>x (meters)</th><th>y (dollars)</th></tr>
<tr><td>0</td><td></td></tr>
<tr><td>50</td><td></td></tr>
<tr><td>100</td><td></td></tr>
<tr><td>150</td><td></td></tr>
<tr><td>200</td><td></td></tr>
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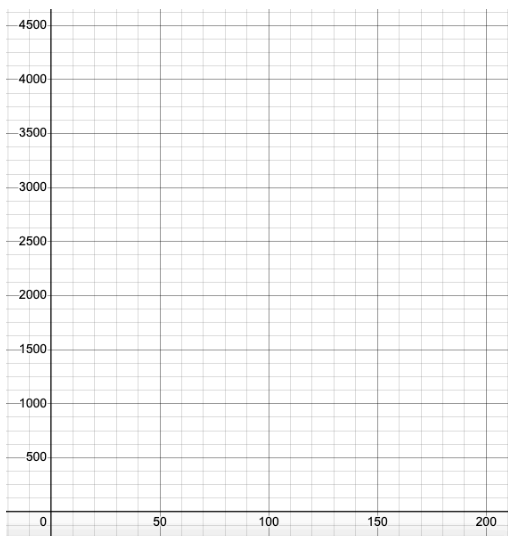

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d) Use the **graph** to estimate the cost of a 70 meter sidewalk.

e) Use the **equation** to **calculate** the cost of a 70 meter sidewalk.